Solution Requirements

Date	15 March 2026
Team ID	xxxxxx
Project Name	AI Learning Support Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	**Not Applicable.** The application does not require user login or authentication	Low
2	Authorization levels	**Not Applicable.** The application has a single user interface with no user roles or permission levels.	Low
3	External interfaces	Integrates with the **Google Gemini API** to generate personalized learning guidance based on the detected emotion and the user's learning problem.	High
4	Transactions	Processes the user's selected	High

	processing	subject and learning problem, performs emotion prediction using **BiLSTM** and **BERT**, generates an AI response, and stores the interaction in a CSV file.	
5	Reporting	Displays emotion analysis, BiLSTM probability chart, BERT prediction, analytics dashboard, previous interaction history, and allows downloading the interaction history as a CSV file.	High
6	Business rules, etc.	Validate that the learning problem is not empty before processing. Detect the primary and secondary emotions, generate personalized AI guidance, store each interaction, and update the analytics dashboard automatically.	High
7	Compliance to laws or regulations	No specific regulatory compliance is implemented. The application processes only user-entered learning queries and stores them locally in CSV format.	Low
8	*Other*	Maintain previous interaction history, display analytics using Plotly charts, and allow users to review earlier learning sessions.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	Emotion prediction, AI response generation, and dashboard updates should complete quickly for a smooth user experience.	Typical response within **5–10 seconds** depending on Gemini API response time.
2	Scalability	Designed for single-user or small-scale educational use with locally stored interaction history.	Supports continuous addition of interaction records without affecting functionality.
3	Security & Data Privacy	Interaction history is stored locally in a CSV file. No authentication or sensitive personal information is collected.	User data remains stored on the local system.
4	Reliability & Availability	The application should consistently load the trained models, process user input, and display results without crashes.	Successful execution of the complete prediction workflow during normal operation.
5	Usability & Accessibility	Provide a simple and responsive Streamlit interface with clear navigation, readable charts, and organized result sections.	Users can submit a problem and view results with minimal steps.